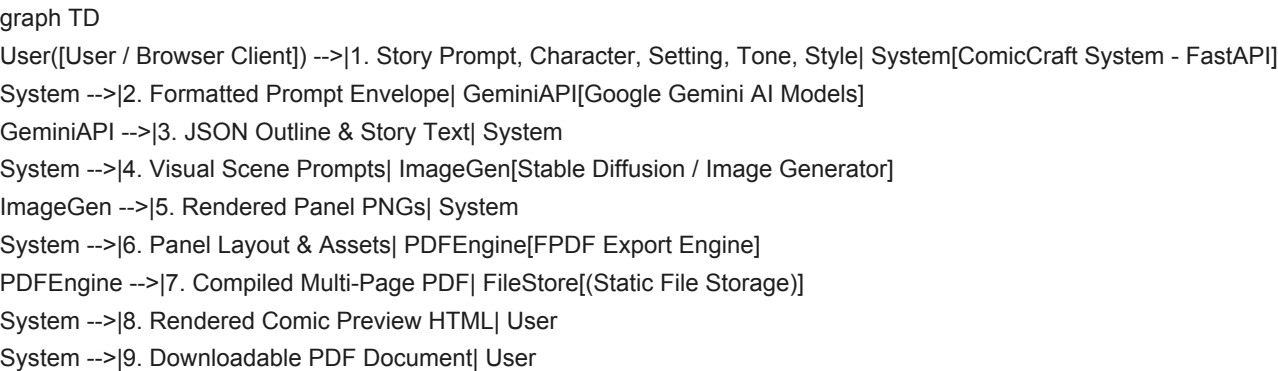


Phase 2: Requirement Analysis

Data Flow Diagram (DFD)

1. DFD Level 0: Context Diagram

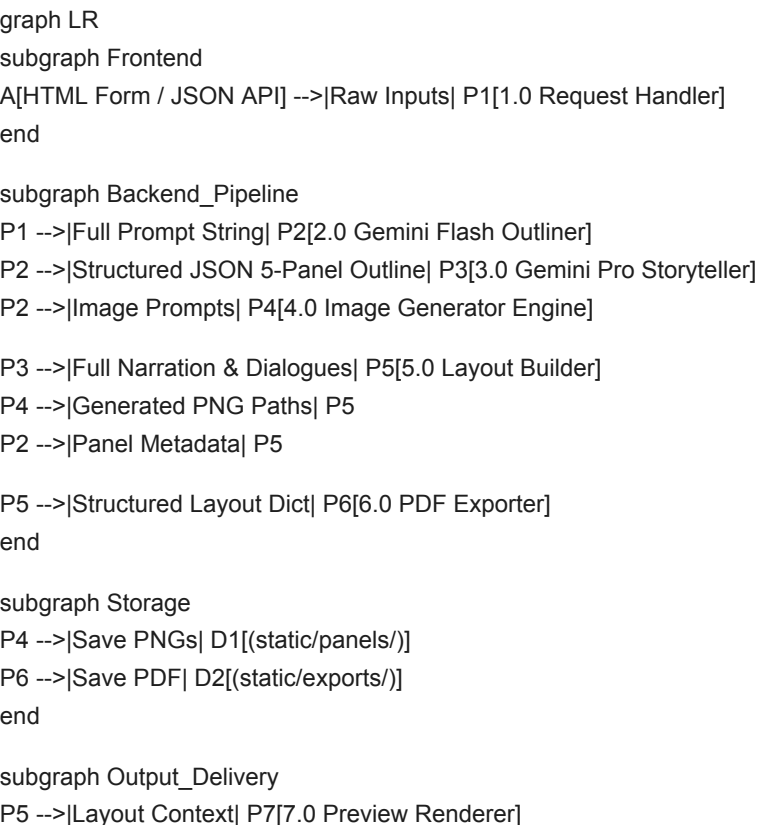
The Level 0 Context Diagram depicts ComicCraft as a single central process interacting with external entities (Users, Google Gemini AI API, Image Generation Engine, and Local File System).



2. DFD Level 1: Subsystem Decomposition

The Level 1 DFD decomposes the system into its primary operational sub-processes:

- 1. Input Ingestion & Validation (routes.py)
- 2. 5-Panel Outline Generation (gemini_flash.py)
- 3. Dialogue & Narration Expansion (gemini_pro.py)
- 4. Concurrent Panel Inking (image_generator.py)
- 5. Layout Assembly & Binding (layout_builder.py)
- 6. PDF Compilation & Storage (exporters.py)
- 7. Client Presentation & Download (comic_preview.html, export_success.html)



P6 -->|PDF Web Path| P7
P7 -->|HTML Response| Client[User Browser]
D2 -->|File Stream| Client
end

3. Data Dictionary

Data Element	Type	Source	Destination	Description
prompt	String	User Form / JSON	generate_outline()	The core story premise or narrative spark.
character_name	String	User Form / JSON	generate_outline()	Name of the primary character.
setting	String	User Form / JSON	generate_outline()	Environmental context (e.g. Forest, Space).
tone	String	User Form / JSON	generate_outline(), generate_story()	Narrative mood (e.g. Dramatic, Funny).
style	String	User Form / JSON	image_generator.py	Visual aesthetic (e.g. Anime, Realistic).
outline	List[Dict]	gemini_flash.py	gemini_pro.py, image_generator.py	5 items containing panel, title, scene_description, image_prompt.
full_story	String	gemini_pro.py	layout_builder.py	Formatted story containing panel titles, dialogues, and captions.
images	List[Str]	image_generator.py	layout_builder.py, static/panels/	File paths to the saved 5 panel illustrations.
layout	List[Dict]	layout_builder.py	save_pdf(), comic_preview.html	Unified collection of panels binding images, titles, descriptions, and story text.
pdf_path	String	exporters.py	comic_preview.html, export_success.html	Absolute and relative path to the generated comic PDF file.